abc atlas programatic access/

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brain science/data & technology

abc atlas access/ https://github.com/AllenInstitute/abc_atlas_access/

- Provides a Python class AbcProjectCache that:
  - downloads data released through the ABC Atlas
  - tracks data release versions
  - tracks what files have been downloaded
  - verifies file integrity
  - provides an easier way of getting at data whether that be an anndata file, cell/gene metadata tables etc.
- Provides a set of tutorial Jupyter notebooks that:
  - Walk the user through how to use the cache object.
  - Show how to interact with the data through example visualizations that can be created with the various data released in the ABC Atlas.
- A Website describing the data and displaying notebooks

abc atlas access website/ https://alleninstitute.github.io/abc_atlas_access/intro.html

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Allen Brain Cell Atlas - Data Access

INTRODUCTION

- Getting started
- Accessing 10x RNA-seq gene expression data
- Using cells selected in the ABC Atlas with Gene Expression data.

DATA

- Mouse whole-brain transcriptomic cell type atlas (Hongkui Zeng)
- Aging Mouse dataset (Hongkui Zeng)
- A molecularly defined and spatially resolved cell atlas of the whole mouse brain (Xiaowei Zhuang)

Allen Brain Cell Atlas - Data Access

The Allen Brain Cell Atlas (ABC Atlas) aims to empower researchers worldwide to explore and analyze multiple whole-brain datasets simultaneously. As the Allen Institute and its collaborators continue to add new modalities, species, and insights to the ABC Atlas, this groundbreaking platform will keep growing, opening up endless possibilities for groundbreaking discoveries and breakthroughs in neuroscience. With the ABC Atlas, researchers everywhere can gain new insights into the brain's complex workings, advancing our understanding of this amazing organ in ways we never thought possible. The ABC Atlas can be accessed at here and is the primary method of interacting with these data. Any questions or issues associated with the ABC Atlas are best directed to the Allen Institute Community Forum. This repository is intended for users who wish to download the ABC Atlas data and processes it locally.

Data associated with the ABC Atlas is hosted on Amazon Web Services (AWS) in an S3 bucket as a AWS Public Dataset, arn:aws:s3:::allen-brain-cell-atlas. No account or login is required for access. ***The purpose of this repo is to provide an overview of the available data, how to download and use it through example use cases.***

The Winter 2024 public beta data release includes:

- **Mouse whole-brain transcriptomic cell type atlas (Hongkui Zeng)**
- **Aging Mouse transcriptomic cell type atlas (Hongkui Zeng)**
- **A molecularly defined and spatially resolved cell atlas of the whole mouse brain (Xiaowei Zhuang)**
- **Human whole-brain transcriptomic cell type atlas (Kimberly Siletti)**

helpful links/

- Python/conda: https://docs.anaconda.com/anaconda/install/
- Jupyter: https://jupyter.org/
- Pandas: https://pandas.pydata.org/docs/
- Anndata: https://anndata.readthedocs.io/en/stable/
- abc_atlas_access:
    - Website
    - Github code repository
    - Notebooks

what's in the demo today/

- Quick run through of the getting_started notebook.
  - Including how to setup the cache and download data.

- Processing of Whole Mouse Brain (WMB) 10X data
  - Extracting genes from the WMB-10X files

- Plotting celltypes and genes expressed in L5 ET CTX Glut in a UMAP

- Plotting genes expressions in a heatmap

- Matching cells selected from an ABC Atlas visualization

demo links/

- Demo files on Github:
  - https://github.com/AllenInstitute/SPOCTT/tree/main/2026_Describe_Your_Neurons_Workshop/abc_atlas_access_demo
- SPOCT CodeOcean Capsule:
  - https://codeocean.allenneuraldynamics.org/capsule/3027664/tree


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thank you/